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The center of an associative crossed module

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Crossed modules of associative algebras are equivalent to internal categories in associative algebras, that is, to strict associative 2-algebras. For a crossed module $\mu\colon \mathsf{M} \to \mathsf{A}$, the multiplication of A makes the associated internal category $\mathbf{Cat}(\mathsf{M},\mathsf{A},\mu)$ a strict semimonoidal category. In general it has no unit, since our algebras need not be unital. We introduce and study a center in this setting, inspired by the Drinfeld center of a monoidal category and by M. Pirashvili's centers of crossed modules of groups and of Lie algebras. Its objects are triples (σ,τ,a), where $a \in \mathsf{A}$ and $\sigma\colon \mathbf{0} \Rightarrow a\cdot(-)$, $\tau\colon \mathbf{0} \Rightarrow (-)\cdot a$ are natural transformations from the zero endofunctor that are compatible with the semimonoidal structure. In other words, each object a comes with coherent null-homotopies of its left and right multiplication functors. The difference $\sigma-\tau\colon (-)\cdot a \Rightarrow a\cdot(-)$ plays the role of a half-braiding, but the pair (σ,τ) retains more information, so the construction is a Drinfeld-type center rather than the Drinfeld center itself. We prove that this center is canonically isomorphic to the internal category of an explicit crossed module $\mathbf{Z}(\mathsf{M},\mathsf{A},\mu)$, built from Mac Lane's bimultiplications. We also show that it carries a natural braiding satisfying the strict hexagon identities. Hence the center is a braided semimonoidal category, corresponding to a braided crossed module. It fits into a crossed square and into a homotopy fibre sequence that relates its homotopy invariants π_0 and π_1 to those of $(\mathsf{M},\mathsf{A},\mu)$. Associative crossed modules form a semi-abelian category, so Huq's categorical notion of center is also available. We compute the Huq center explicitly. We then show that it embeds into our center through a morphism that is an isomorphism on π_1 and injective on π_0, but is not a weak equivalence in general. Finally, we compare our center with Pirashvili's center of Lie crossed modules via the (braided) Liezation functors. We obtain a canonical comparison morphism, which preserves braidings under an injectivity condition.

Keywords: Category theory; Crossed modules; Center
Subject area: Crossed modules and category theory

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